ABEEHA MIRZA BAIG



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INTRODUCTION

Currently pursuing Master's in Data Sciences at NED University of Engineering and Technology (NEDUET), I am an AI/ML Engineer with two years of hands-on experience in artificial intelligence. My expertise lies in natural language processing (NLP), deep learning, text preprocessing, AI agents, retrieval-augmented generation (RAG), large language models (LLMs), and developing APIs with Flask. Driven by a passion for harnessing advanced AI techniques, I am committed to delivering impactful solutions and advancing the boundaries of data-driven innovation.

WORK EXPERIENCE

Machine Learning Engineer at PixelEdge Solutions (Pvt.) Limited, Karachi, Pakistan, October 2023 – Present

- Processed financial and technical files by applying NLP techniques, including scraping and preprocessing, to extract valuable insights and generate formatted outputs.
- Implemented various forms of Retrieval-Augmented Generation (RAG)—modular, basic, and advanced—using techniques like HyDE, ingestion techniques, re-ranking., etc to enhance chatbot functionality.
- Designed and deployed AI agents using LangGraph and Langchain to handle multi-step queries and search processes for healthcare and financial products.
- Applied Pandas agent and SQL Agent using langchain's framework using python REPL tool to process Financial data and gain insights from it through natural language input.
- Performed prompt engineering and integrated large language models (LLMs) such as DeepSeek, LLaMA, OpenAI, Gemini, and Claude to address complex problems, analyzing results to deliver effective solutions.
- Developed Flask APIs and Fast APIs with robust error handling to seamlessly integrate solutions into development and production environments.
- Fine-tuned GPT-2 for a next-word prediction model using healthcare data, optimizing performance for domain-specific applications.

Machine Learning Intern at Techno-Hacks Edu Tech, Karachi, Pakistan, August 2023 – September 2023

- Gained hands-on experience in implementing machine learning principles to solve real-world problems.
- Conducted predictive analysis to forecast trends and outcomes, leveraging various ML algorithms and techniques.
- Performed classification tasks to categorize data based on specific criteria, enhancing the accuracy and efficiency of data processing.
- Developed a strong foundation in basic ML concepts, applying theoretical knowledge to practical applications.

EDUCATION

1. Master of Science (Data Sciences) – NED University of Engineering and Technology

January 2025 - ongoing (Courses: Statistics and probability for DS, Machine Learning, Deep Learning, Artificial neural networks, Data mining, Data visualization, OOP, advanced database techniques, tools and techniques for data sciences, generative AI)

COURSES AND CERTIFICATIONS

- **AI Agents** | Hugging-Face
- **Generative AI with LLMs** | Coursera
- **OCI Generative AI professional Certification** | Oracle University
- **Building Recommendation Systems using Machine Learning** | Great Learning Academy
- **Building Systems with Chat-GPT API and GEMINI API** | DeepLearning.AI
- **Python Programming** | Open weaver

PERSONAL PROJECTS

Financial data bot Using Langchain's Pandas agent

https://github.com/abeeha-baig/financial-chatbot

Developed an intelligent Financial Data Chatbot that leverages OpenAI's language models and the Pandas Agent framework to enable seamless interaction with CSV-based financial datasets through natural language queries.

Lifestyle Agent using Lang-Chain and Fire-Crawl

https://github.com/abeeha-baig/Lifestyle-coach-agent.git

It helps you explore wellness, productivity, and self-improvement tips by scraping articles from trusted sources like Medium, Healthline & more (via Firecrawl). Summarizing them with GPT-4o-mini for quick takeaways. Built using Python, Streamlit, LangChain, and secure API handling.

Modular RAG based chatbot for Oracle database Security Q/A
(https://github.com/abeeha-baig/RAG-ODS.git**)**

This project is a Retrieval-Augmented Generation (RAG) Q&A System built using Streamlit and LangChain. It allows users to interact with a conversational AI that can retrieve relevant information from embedded documents (PDFs) and generate coherent responses..

Next Word Prediction by Finetuning GPT-2 model
(https://github.com/abeeha-baig/Next-Word-Prediction.git *)

This project involved the finetuning of gpt-2 with some data about health, and producing the next word

SKILLS

- Natural Language Processing (NLP): Transformers, NLTK, feature extraction, text preprocessing, Prompt engineering, RAG.
- Agentic AI with Langchain, & CrewAI
- Machine Learning Algorithms: Regression, Classification, Clustering, Neural Networks
- Data Analysis: Exploratory Data Analysis (EDA), Model Evaluation
- Programming Languages: Python, SQL
- Data Visualization: Matplotlib, Seaborn
- Tools: Jupyter Notebook, VS Code, Git & GitHub, PyTorch, Scikit-learn
- Communication: Strong verbal and written communication skills
- Problem Solving: Analytical thinker with a knack for solving complex problems
- Adaptability: Quick learner and adaptable to new technologies and methodologies
- Time Management: Efficiently manage multiple tasks and projects